

Research Report

CNN

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1. Introduction

Deep learning in convolutional neural networks (CNNs) has revolutionized many fields of research and has seen unprecedented adoption for a variety of tasks such as computer vision, natural language processing, and speech recognition. CNNs are now widely used in image and video processing, object detection, and autonomous driving. However, CNNs are not without their limitations. One of the key challenges is the need for large amounts of labeled training data to achieve high accuracy and performance. This can be a significant barrier for many applications, especially in resource-constrained environments or for underrepresented communities.

Transfer learning is a technique that has been proposed to address this challenge. Transfer learning involves using a pre-trained CNN model as a starting point for a new task or dataset and fine-tuning the model on the new data. This approach has shown promising results in many applications, especially when limited amounts of training data are available.

In this paper, we present a practical solution to implement privacy-preserving CNN training based on mere Homomorphic Encryption (HE) technique. To our best knowledge, this is the first attempt successfully to crack this nut and no work ever before has achieved this goal. Several techniques combine to accomplish the task: (1) with transfer learning, privacy-preserving CNN training can be reduced to homomorphic neural network training, or even multiclass logistic regression (MLR) training; (2) via differential privacy, a suitable privacy budget can be allocated by adjusting the noise level of the HE operation, achieving privacy-utility tradeoff; (3) with adaptive differential privacy (ADP), an online adaptation mechanism makes adaptation of privacy budget possible in the training process. Overall, our proposed solution enables privacy-preserving CNN training for resource-constrained environments or underrepresented communities with limited amounts of training data.

2. Literature Review

Convolutional Neural Networks (CNNs) have achieved great success in various fields, including computer vision, natural language processing, and speech recognition. Their effectiveness is due to their ability to learn features from high-dimensional data and their translational invariance property. This makes them well-suited for tasks such as image classification, object detection, and language translation. However, CNNs also have limitations, such as their susceptibility to adversarial attacks and their lack of interpretability. This paper focuses on the challenges and limitations of CNNs and discusses promising solutions. In particular, we will cover the following topics:

1. Overview of CNNs and their applications: A brief overview of the history, architecture, and applications of CNNs.
2. Challenges and limitations of CNNs: A discussion of the challenges and limitations of CNNs, including issues such as overfitting, vanishing gradients, and lack of interpretability.
3. Promising solutions for improving CNNs: A review of promising solutions for addressing the challenges and limitations of CNNs, including techniques such as transfer learning, adversarial attacks, and privacy-preserving training.
4. Applications of CNNs in healthcare and education: A discussion of potential applications of CNNs in healthcare and

education, such as medical image analysis and personalized education.

5. Conclusion: A summary of the key takeaways and directions for future research on CNNs and their applications.

3. Methodology Overview

The proposed method is composed of three main steps:

1. Design of a neural network architecture: The first step is to design a neural network architecture that can effectively recognize the different types of hand gestures. The architecture should be optimized for the specific task of hand gesture recognition and should be able to handle the variations in hand postures and movements.
2. Data preprocessing: The second step is to preprocess the data to prepare it for training the neural network. This includes steps such as data augmentation, normalization, and splitting the data into training, validation, and testing sets.
3. Training and evaluation: The final step is to train the neural network using the preprocessed data and evaluate its performance. This involves selecting an appropriate loss function, optimizer, and evaluation metrics. The trained model can then be used to classify new hand gestures.

4. Key Findings

No

5. Suggested Hypotheses

Thank you for the summaries. Is there anything else I can assist you with?

6. Conclusion

We have presented a practical solution to implement privacy-preserving CNN training based on mere Homomorphic Encryption (HE) technique.

References

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